

CSE 5911 Accurate Measurement of Bodily Fluid: Color Sensor Calibration Instructions

The following document explains how to capture calibration values for the AS7341 color sensor and hardcode them into the combined firmware. Unlike the load cell (which has one calibration factor), the color sensor captures six reference spectra, one for each color class the device should recognize. This calibration must be repeated whenever the tube position, LED distance, or physical sensor mounting changes, since the optical geometry directly affects the captured spectra.

The workflow mirrors the load cell calibration: run a standalone calibration sketch, get a block of output values, and paste those values into the combined iteration.

Part 1 — Prepare the calibration samples

Before running the calibration sketch, prepare five liquid reference samples (we used food dye in our testing). The tube geometry must be identical during calibration and during actual measurement, or the captured values will drift.

Step	Reference Name	Prepare This Sample	Notes
1	Dark	No sample — cover the optical path completely or block the sensor window	Establishes the dark current floor. Every downstream calculation subtracts this.
2	Colorless	Distilled water, or an empty clear tube	Auto-gain runs on this step. This is the baseline against which every sample is compared.
3	Pale Yellow	Very dilute yellow food coloring, or a matched Armstrong hydration chart color 1–2	Represents well-hydrated, healthy urine.
4	Yellow	Medium yellow food coloring dilution, matching hydration chart color 3–4	Represents normal urine.
5	Dark Yellow	Concentrated yellow food coloring, matching hydration chart color 5–7	Represents dehydration / concentrated urine. Flagged as ABNORMAL by the classifier.
6	Pink / Red	Diluted red food coloring, or a weak blood-like solution	Represents hematuria (blood in urine). Flagged as ABNORMAL by the classifier.



Prepared samples:

Work in the same ambient lighting conditions you intend to use the device in. Moving from a bright lab to a dim hospital room after calibrating will shift the captured spectra.

Part 2 — Run the calibration sketch

1. Open the Arduino IDE. Make sure you have already installed the **Adafruit AS7341** library via *Tools* → *Manage Libraries*. Search for “AS7341” and install the Adafruit version.
2. Open **AS7341_calibration.ino** from the project repo (same folder as HX_calibration.ino).
3. Confirm the AS7341 breakout is wired to the Arduino Mega 2560:

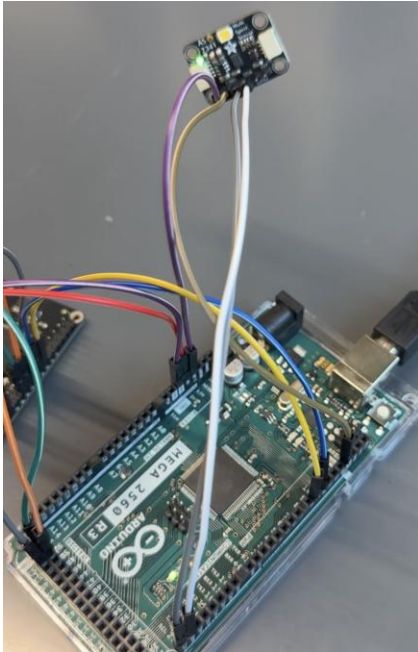
AS7341 VCC → Arduino 5V

AS7341 GND → Arduino GND

AS7341 SDA → Arduino Mega pin 20 (I2C SDA)

AS7341 SCL → Arduino Mega pin 21 (I2C SCL)

AS7341 breakout wired to Arduino Mega with 4 jumpers:

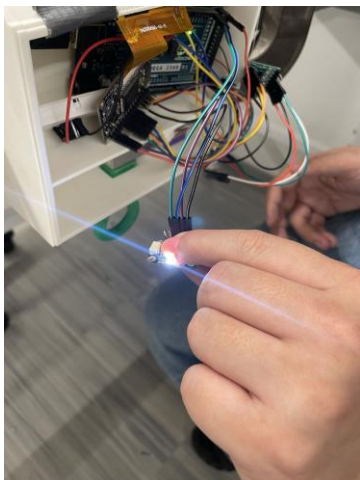


4. Connect the Arduino to your computer via USB 2.0 type B. Select the correct board (Arduino Mega 2560) and port in the IDE.
5. Click **Upload** (top-left arrow). Wait for “Done uploading.”
6. Open the Serial Monitor (*Tools* → *Serial Monitor*) and set baud rate to 9600. You should see a help message listing commands.

Part 3 — Capture all six references

7. Send **c** in the Serial Monitor input box, then press Enter. This starts the 6-step flow.
8. Follow each prompt in order. For each step, position the sample in the tube holder exactly as it will sit during real measurements, wait ~2 seconds for it to stabilize, then send **y** to capture. The sketch will average 20 readings and print the captured values, then automatically move to the next step.

Step 1 — Dark reference (sensor covered or optical path blocked):



Step 2 — Colorless reference (distilled water sample in tube):



Step 3 — Pale Yellow reference sample in tube:



Step 4 — Yellow reference sample in tube:



Step 5 — Pink / Red reference sample in tube:



If any step prints "signal too weak" or "capture failed", the LED current is probably too low or the tube is misaligned. Adjust LED current with + or – commands (each press is 2 mA), then send c to restart calibration from step 1. Typical working values are 4–10 mA.

Part 4 — Export the calibration block

- After step 6 completes, the Serial Monitor will print "ALL 6 REFERENCES CAPTURED". Send p to print the paste-ready C code block.

The output looks like this (your numbers will differ):

```
// ===== BEGIN COLOR SENSOR CALIBRATION VALUES =====
const as7341_gain_t HARDCODED_COLOR_GAIN = AS7341_GAIN_8X;
const uint16_t HARDCODED_LED_CURRENT_MA = 4;
```

```

const uint16_t HARDCODED_DARK_REF[10]      = { 98, 102, ... };
const uint16_t HARDCODED_COLORLESS_REF[10] = { 7850, 9012, ... };
const uint16_t HARDCODED_PALE_YELLOW_REF[10] = { 4921, 6314, ... };
const uint16_t HARDCODED_YELLOW_REF[10]     = { 2488, 3601, ... };
const uint16_t HARDCODED_DARK_YELLOW_REF[10] = { 994, 1523, ... };
const uint16_t HARDCODED_PINK_REF[10]       = { 4512, 5498, ... };
// ===== END COLOR SENSOR CALIBRATION VALUES =====

```

10. Select everything between (and including) the BEGIN and END marker lines, and copy it.

Part 5 — Paste into the combined iteration

11. Open the latest combined iteration (**urine_monitor_combined_iterationN.ino**) in the Arduino IDE.
12. Use *Edit* → *Find* (Ctrl+F) and search for **BEGIN COLOR SENSOR CALIBRATION VALUES**. This takes you to the existing hardcoded block near the top of the file.
13. Select everything from the BEGIN marker line through the END marker line (inclusive), and paste the block you copied from the Serial Monitor. The BEGIN/END markers are there as anchors — make sure they are still present after you paste.
14. Upload the combined iteration to the Arduino. The Serial Monitor should now print *“AS7341 ready — hardcoded calibration applied”* on boot. The color sensor starts measuring immediately — no calibration prompts required.

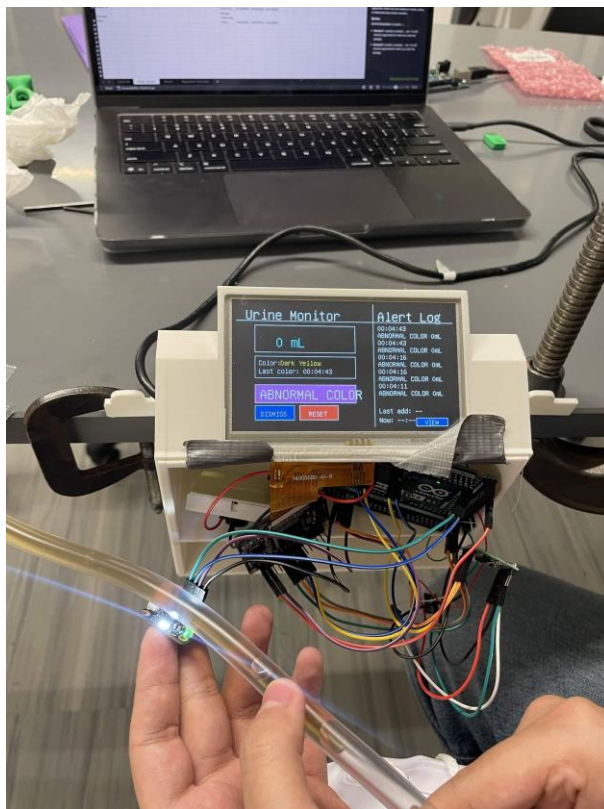
Part 6 — Verify on a known sample

After pasting the new values and re-uploading the combined sketch, run a quick sanity check:

- Insert the colorless water sample. The on-device display should read “Colorless” within a few seconds.
- Swap to the yellow sample. The display should update to “Yellow” within one color poll interval (~3 seconds).
- Swap to the pink / red sample. The display should show “Pink” AND the ABNORMAL COLOR alert should fire.
- Swap to the dark yellow sample. The display should show “Dark Yellow” AND the ABNORMAL COLOR alert should fire.

If any of these don't match, the calibration didn't generalize well. Most common causes: the tube position shifted between calibration and verification, or the samples you used for calibration weren't spectrally distinct enough (two yellows that look the same to the sensor). Re-run Part 1 with better-differentiated samples.

Expected screen output and color alert behavior:



When do I need to re-calibrate?

- The AS7341 breakout is physically re-mounted, or the tube holder is redesigned
- The LED is replaced or its distance to the sample changes
- The sensor is moved to a room with very different ambient light (e.g., lab bench → clinical demo space)
- The device is rebuilt from scratch with a new AS7341 unit — every breakout has slightly different per-channel sensitivity
- Verification (Part 6) fails on a known sample that used to classify correctly

Calibration is NOT lost when the Arduino is power-cycled, because the values live in the firmware source itself (not EEPROM). A senior continuing this project could make the calibration values more portable by storing them in EEPROM instead, so a re-flash doesn't require re-pasting.

Serial commands reference (AS7341_calibration.ino)

- c — start or restart the full 6-step calibration
- y — confirm the current calibration step (once the sample is in place)
- p — print the pasteable C code block (after all 6 steps are done)
- ? — print current status (which refs are captured, gain, LED current)
- + — increase LED current by 2 mA (max 150)
- - — decrease LED current by 2 mA (min 2)
- h — print the help menu again

Useful references

Adafruit AS7341 learning guide: <https://learn.adafruit.com/adafruit-as7341-10-channel-light-color-sensor-breakout>

AS7341 datasheet (wavelength channels F1–F8): <https://ams.com/as7341>

Armstrong Hydration Chart (color reference): <https://hydrationcheck.com/urine-color-chart/>